

The 24–600 Spectral Bridge

A selective $\lambda=12$ eigenspace embedding from five 24-cell cosets into the 600-cell

Status: Reproducible mathematical note / exact computational certificate. Not peer-reviewed.

Reproducibility. The formal claims of this note — Theorem 1 (the $\lambda=12$ lift identity); the $\dim E_{V_{600}}(\lambda)$ entries for $\lambda \in \{0, 4, 8, 10, 12\}$ used in the selectivity table of Section 4; Lemma 4 ($\{\pm 1\}$ -triviality); Lemma 5 ($2I$ -invariance of $E_{\text{lifted}}, E_{\text{residual}}$); Theorem 6 (integer A_5 class characters); and Corollary 7 (irrep multiplicities) — are all certified by exact rational arithmetic in the accompanying repository (integer Laplacians, $\mathbb{Q}$ -rational nullspace bases, exact zero residuals; the A_5 character projection uses exact $\mathbb{Q}(\sqrt{5})$ character-table entries). The numerical quantities reported in Remark 2 and the “max-residual” column of the selectivity table are floating-point cross-checks only. One-command reproduction and the frozen output artifacts are described in Section 7.

Abstract

The 600-cell is the regular 4-polytope whose 120 vertices form the binary icosahedral group $2I$. The binary tetrahedral group $2T \subset 2I$ has index 5; its right cosets partition V_{600} into five disjoint copies of the 24-cell. Each coset carries a $d=1$ (nearest-neighbour) graph isomorphic to the 24-cell edge graph (8-regular); on the full vertex set the $d=1$ graph is the 600-cell edge graph (12-regular).

We prove that the 2-dimensional $\lambda=12$ eigenspace of each local 24-cell Laplacian zero-extends into the 25-dimensional $\lambda=12$ eigenspace of the full 600-cell Laplacian. The identity is established *exactly over* $\mathbb{Q}$: from the local $\mathbb{Q}$ -rational nullspace of each $L_C - 12I$ (an integer matrix), every zero-extended basis vector is verified to lie in the kernel of $L_{V_{600}} - 12I$ by exact rational matrix multiplication. A floating-point cross-check independently gives residual $\leq 6.2 \times 10^{-15}$.

Among the five local 24-cell Laplacian eigenvalues $\{0, 4, 8, 10, 12\}$, $\lambda=12$ is the only one that gives a full zero-extension into a corresponding global 600-cell eigenspace. The $\lambda=0$ case contributes exactly one direction (the global constant, reconstructed from the symmetric sum of the five coset indicators) of the 5-dimensional lift image; the local eigenvalues $\lambda \in \{4, 8, 10\}$ are not eigenvalues of the 600-cell Laplacian at all, so the question is vacuous there.

Under the induced $A_5 = 2I/\{\pm 1\}$ action on the five cosets, $E_{V_{600}}(12) = E_{\text{lifted}} \oplus E_{\text{residual}}$ with $E_{\text{lifted}} \cong 2 \cdot Y_5$, $E_{\text{residual}} \cong 3 \cdot Y_5$, $E_{V_{600}}(12) \cong 5 \cdot Y_5$, where Y_5 is the 5-dimensional irreducible representation of A_5 .

1 Introduction

The 600-cell is the regular 4-polytope $\{3, 3, 5\}$ with 120 vertices, 720 edges and icosahedral symmetry. Its vertex set is naturally identified with the binary icosahedral group $2I \subset \text{Sp}(1)$. The 24-cell is the regular 4-polytope $\{3, 4, 3\}$ with 24 vertices, 96 edges and F_4 -symmetry; its vertex set is the binary tetrahedral group $2T$.

It is classical that $2T$ has index 5 in $2I$ and that the five cosets of $2T$ in $2I$ form five inscribed 24-cells whose vertex sets partition V_{600} [Coxeter 1973, Conway–Sloane 1999]. An explicit machine-verified treatment within the present programme is [Paper XXXV]: the Schläfli decomposition is proved in section `sec:bio:schlafl`i (Theorem `thm:schlafl`i; supporting script `papers/cascade-derivation/scripts/verify_schlafl`i_via_2T.py), and

the A_5 origin of the coset action is proved in section `sec:gr:lorentz` (Theorem `thm:A5action`; script `papers/cascade-derivation/scripts/coset_action_2I_on_5_skeletons.py`).

The question this note answers is sharper: *does the spectral structure of each local 24-cell graph embed exactly into the spectral structure of the global 600-cell graph?* The answer is “yes, but only at $\lambda=12$ in full”. Full zero-extension to a global λ -eigenspace occurs only at $\lambda=12$; at $\lambda=0$ exactly one symmetric combination (the sum of the five coset indicators, equal to the global constant) lands in the global eigenspace, while at every other tested λ the lift image intersects the global eigenspace only at 0.

Why the comparison is non-trivial. The local 24-cell Laplacian L_C and the global 600-cell Laplacian $L_{V_{600}}$ are built from *different* graphs: L_C uses the 24-cell edge graph on the 24 vertices of a single coset (edges at squared distance 1), while $L_{V_{600}}$ uses the 600-cell edge graph on all 120 vertices (edges at squared distance $(3 - \sqrt{5})/2 < 1$). These graphs are not subgraphs of one another; in fact *no* edge of L_C is an edge of $L_{V_{600}}$ (the local 24-cell edge length is the next distance shell up of V_{600}). Consequently, for a function v supported on a coset C_i , the value $(L_{V_{600}}v)(u)$ at a vertex $u \notin C_i$ is in general non-zero — it is minus the sum of v over the global neighbours of u that happen to lie in C_i — so the zero-extension of v is generically *not* a $L_{V_{600}}$ -eigenvector. The content of Theorem 1 is that this off-coset coupling cancels *identically* on the local $\lambda=12$ eigenspace — a finite-rank linear condition that turns out to be satisfied — and the content of the selectivity table is that no other local eigenspace satisfies it. This is what makes the $\lambda=12$ channel a genuine local-to-global bridge rather than a trivial restriction-and-extension.

2 Construction

2.1 V_{600} as $2I$

We work in exact $\mathbb{Q}(\sqrt{5})$ arithmetic. The 120 vertices of V_{600} are:

- 8 vertices $\pm e_i$, $i = 0, \dots, 3$;
- 16 vertices $(\pm 1, \pm 1, \pm 1, \pm 1)/2$;
- 96 vertices: even permutations of $(0, \pm 1/(2\varphi), \pm 1/2, \pm \varphi/2)$, $\varphi = (1 + \sqrt{5})/2$.

These 120 unit quaternions form $2I$ under quaternion multiplication; the construction (exact $\mathbb{Q}(\sqrt{5})$ arithmetic, all 120 vertices) lives in the package `vfd_v600` [`V600-programme`]. Closure of the vertex set under multiplication is exercised end-to-end by the A_5 certificate of Section 5 (which enumerates every $h \in 2I$ and looks up $vh \in 2I$ for every $v \in 2I$); any non-closure would manifest as a missing index lookup and abort the certificate.

2.2 V_{24} as $2T = \sigma$ -fixed subset

The Galois twist $\sigma: \sqrt{5} \mapsto -\sqrt{5}$ acts component-wise on $\mathbb{Q}(\sqrt{5})$. Its fixed subset in V_{600} is the 24-element binary tetrahedral group $V_{24} = 2T$: the 8 unit vectors $\pm e_i$ together with the 16 half-integer vectors. Verification is in [`V600-programme`] (`vfd_v600.sigma`).

2.3 Right cosets of $2T$ in $2I$ and the A_5 action

Convention. We work with the *right cosets* $2T \cdot v$ in $2I$; equivalently, with the orbits of left multiplication by $2T$ on $2I$. Left and right cosets give different partitions of $2I$ (since $2T$ is not normal in $2I$), but both partition the 120-vertex set into five sets of 24, both contain $2T$ itself as the identity coset, and both produce isomorphic 24-cell edge graphs on each block by isometry. The right-coset convention is the one used throughout the keystone-artifact code; the left-coset version is treated in [Paper XXXV] (section `sec:bio:schlafl`). The two conventions are

related by the involution $g \mapsto g^{-1}$ of $2I$, which is a bijection between left and right cosets fixing the identity coset $2T$; hence the spectral-bridge claims of this note transfer between conventions.

Cosets. The five right cosets of $2T$ in $2I$ are

- one coset $C_0 = 2T$ itself (the σ -fixed orbit);
- four cosets $C_1, \dots, C_4$: the four $2T$ -orbits on the 96 σ -mobile vertices.

Each $|C_i| = 24$, and $\bigsqcup C_i = V_{600}$ [Paper XXXV, V_{600} -programme].

Action on cosets. $2I$ acts on the five right cosets by right multiplication: $(2T \cdot v) \cdot g = 2T \cdot (vg)$. The kernel of this *coset-level* action contains $\{\pm 1\}$ because $-1 \in 2T$ (since $2T$ contains $\pm e_0 = \pm 1$ as one of its 24 elements), so for any right coset $2Tv$ we have $2Tv \cdot (-1) = 2T(-v) = (-2T)v = 2Tv$. By the parallel left-coset result in [Paper XXXV] (proved at all 120 elements of $2I$ by `papers/cascade-derivation/scripts/coset_action_2I_on_5_skeletons.py`) combined with the involution $g \mapsto g^{-1}$ which exchanges left- and right-coset families, the kernel of this action is exactly $\{\pm 1\}$ and the image on five letters is exactly $A_5 = 2I/\{\pm 1\}$, acting transitively.

Vertex action vs. coset action. By contrast, right multiplication on *vertices* of V_{600} has trivial kernel: $v \cdot g = v$ for some v forces $g = 1$. When we use the term “ A_5 action” on a subspace $E \subset \mathbb{R}^{V_{600}}$ below, we mean the action induced on E by right multiplication of vertices, restricted to subspaces on which -1 acts trivially (so that the linear action descends through $\{\pm 1\}$ to A_5). Whether -1 acts trivially on a given $E \subset \mathbb{R}^{V_{600}}$ is a non-trivial fact specific to E ; we prove it for $E = E_{V_{600}}(12)$ in Section 5 (Lemma 4).

Object	$ V $	$ E $	Graph (= shortest non-zero distance shell)
V_{600}	120	720	shortest- V_{600} shell, squared distance $(3 - \sqrt{5})/2$, 12-regular
$2T$ -coset (any)	24	96	shortest intra-coset shell, squared distance 1, 8-regular

The two distance shells are distinct: *no* pair within a single coset has separation $(3 - \sqrt{5})/2$, and conversely no V_{600} -edge joins two vertices of the same coset. Throughout this note, “ $d=1$ graph” is shorthand for “shell-1 (nearest-neighbour) graph at the shortest non-zero distance of whichever vertex set we are working on”; the squared distances are different for V_{600} and a coset, as the table makes explicit.

2.4 The two $d=1$ shells

Inside $\mathbb{Q}(\sqrt{5})$, there are 8 distinct non-zero squared distances between pairs in V_{600} . The shortest is

$$d_{V_{600}}^2 = \frac{3-\sqrt{5}}{2} \approx 0.382,$$

which defines the 600-cell edge graph (12-regular, 720 edges). Inside each right coset C_i , the shortest non-zero squared distance between members is the next shell up:

$$d_C^2 = 1,$$

which on C_i realises the standard 24-cell edge graph (8-regular, 96 edges). The two distances are distinct: *no* pair within a single coset has separation $d_{V_{600}}^2$.

These two distinct shells underlie the two Laplacians compared in this note.

2.5 Laplacians

We use the unweighted combinatorial Laplacian $L = D - A$. Both graphs are regular and have integer 0/1 adjacency matrices and integer regular degree, so the Laplacians are integer matrices. Numerical diagonalisation (the local spectrum agrees with the floating-point computation in [Paper V], `papers/paper-v/scripts/run_polytope_uniqueness_narrow.py`; every integer-eigenvalue multiplicity in both spectra below is additionally certified by exact $\mathbb{Q}$ -rational SymPy nullspace of $L_{V_{600}} - \lambda I$ and $L_C - \lambda I$ at the specific integer λ values shown in the spectra, via `closure_transform_engine.keystone.exact_integer_spectrum_check`):

$$\text{spec } L_C = \{0^1, 4^4, 8^9, 10^8, 12^2\},$$

$$\text{spec } L_{V_{600}} = \{0^1, (\approx 2.29)^4, (\approx 5.53)^9, 9^{16}, 12^{25}, 14^{36}, (\approx 14.47)^9, 15^{16}, (\approx 15.71)^4\},$$

in multiset notation λ^m ; the four non-integer eigenvalues take values in $\mathbb{Q}(\sqrt{5})$. The integer eigenvalues of $L_{V_{600}}$ (including the multiplicity $\dim E_{V_{600}}(12) = 25$ at $\lambda = 12$ used in Corollary 3 and Section 5) coincide with the $\mathbb{Q}$ -rational kernel dimensions computed by the SymPy nullspace routine in `closure_transform_engine.keystone`, so $\dim E_{V_{600}}(12)$ is itself an exact rational-arithmetic spectral fact.

3 The $\lambda=12$ lift identity

For a coset C_i , define the zero-extension map

$$\text{lift}_i: \mathbb{R}^{24} \rightarrow \mathbb{R}^{120}, \quad (\text{lift}_i u)(v) = \begin{cases} u(v) & v \in C_i, \\ 0 & v \notin C_i. \end{cases}$$

Theorem 1 ($\lambda=12$ lift identity). *For every coset C_i ,*

$$\text{lift}_i(E_{C_i}(12)) \subset E_{V_{600}}(12).$$

Proof by exact rational certificate. Since the local Laplacian L_C has integer entries and 12 is an integer eigenvalue, $E_{C_i}(12) = \ker(L_C - 12I)$ is a $\mathbb{Q}$ -rational subspace of $\mathbb{Q}^{24}$, and $L_{V_{600}}$ likewise has integer entries. The accompanying repository produces a checkable certificate consisting of: (i) the integer matrices $L_C - 12I$ and $L_{V_{600}} - 12I$; (ii) a $\mathbb{Q}$ -rational basis $\{v_1, v_2\}$ of each $E_{C_i}(12) = \ker(L_C - 12I)$, obtained by exact Gaussian elimination over $\mathbb{Q}$; (iii) for each v_k , the vector $(L_{V_{600}} - 12I)\text{lift}_i(v_k) \in \mathbb{Q}^{120}$, computed by exact rational matrix multiplication, with every one of its 120 coordinates equal to 0. Item (iii) is the certificate of $L_{V_{600}}\text{lift}_i(v_k) = 12\text{lift}_i(v_k)$ in $\mathbb{Q}^{120}$, for both $k = 1, 2$ and all five cosets i ; no floating-point arithmetic enters. The certificate is regenerated and re-checked by `closure_transform_engine.keystone.exact_certify_lambda12_lift` and asserted by the test suite. $\square$

Remark 2 (Floating-point cross-check). Independent floating-point diagonalisation of $L_{V_{600}}$ and each L_C via `numpy.linalg.eigh`, followed by zero-extension of each basis vector v of $E_{C_i}(12)$ and evaluation of $\|L_{V_{600}}v_{\text{lifted}} - 12v_{\text{lifted}}\|$, gives maximum residual $\leq 6.2 \times 10^{-15}$ across all 10 lifted basis vectors. This is consistent with Theorem 1 at the level of numerical precision.

Corollary 3. *Setting $E_{\text{lifted}} := \bigoplus_i \text{lift}_i(E_{C_i}(12))$, the five lifted local eigenspaces are linearly independent (their supports are the disjoint cosets), so $\dim E_{\text{lifted}} = 5 \cdot 2 = 10$. Since $\dim E_{V_{600}}(12) = 25$, the subspace $E_{\text{lifted}} \subset E_{V_{600}}(12)$ is proper.*

4 Selectivity

We test whether the analogue of Theorem 1 holds for other eigenvalues λ of L_C . The natural measure is the *intersection dimension*

$$\mathfrak{s}(\lambda) := \dim \left(\text{Im}(\text{lift})|_{\lambda} \cap E_{V_{600}}(\lambda) \right),$$

where $\text{Im}(\text{lift})|_{\lambda} = \sum_i \text{lift}_i(E_{C_i}(\lambda)) \subset \mathbb{R}^{120}$. A “full” lift would have $\mathfrak{s}(\lambda)$ equal to the total lift-image dimension; a generic eigenvalue has $\mathfrak{s}(\lambda) = 0$.

local λ	lift-image dim	$\dim E_{V_{600}}(\lambda)$	$\mathfrak{s}(\lambda)$	verdict
0	5	1	1	PARTIAL — only global constant
4	20	0	0	$E_{V_{600}}(\lambda) = \{0\}$
8	45	0	0	$E_{V_{600}}(\lambda) = \{0\}$
10	40	0	0	$E_{V_{600}}(\lambda) = \{0\}$
12	10	25	10	FULL

The PARTIAL row at $\lambda=0$ is the substantive control: here $E_{V_{600}}(0)$ is one-dimensional (the global constant), and one might ask whether the lift hits it. It does — but only through the symmetric combination: $\mathbf{1}_{V_{600}}$ is the sum of the five coset indicators, each a local $\lambda=0$ eigenvector, so exactly one direction of the 5-dimensional lift image lands in $E_{V_{600}}(0)$. No single-coset $\lambda=0$ lift lies in $E_{V_{600}}(0)$; only the A_5 -invariant combination does. This is the sense in which even the one eigenvalue that the lift “partly” reaches, it reaches only in the symmetry-forced way.

For $\lambda \in \{4, 8, 10\}$ the control is weaker: these are simply not eigenvalues of $L_{V_{600}}$ at all. The exact $\mathbb{Q}$ -rational symplectic nullspace of $L_{V_{600}} - \lambda I$ has dimension 0 for each of $\lambda \in \{4, 8, 10\}$ (certified by `exact_integer_spectrum_check`), so $E_{V_{600}}(\lambda) = \{0\}$ and $\mathfrak{s}(\lambda) = 0$ holds for trivial dimensional reasons. We list these rows for completeness — and the floating-point lift residuals at these λ (3.16 to 9.38, far above machine precision) confirm the lifted vectors are nowhere near an approximate λ -eigenfunction of $L_{V_{600}}$ — but we do not claim they constitute a strong falsification test in the way the $\lambda=0$ row does.

Hence among the five eigenvalues of L_C , only $\lambda=12$ satisfies the lift identity in full; at $\lambda=0$ the lift is partial in the single symmetry-forced way; and at the remaining λ the question is empty for spectral reasons.

5 Direct-sum decomposition and A_5 structure

Inside $E_{V_{600}}(12)$, the lifted subspace E_{lifted} (dim 10) has an orthogonal complement

$$E_{\text{residual}} := E_{V_{600}}(12) \ominus E_{\text{lifted}}, \quad \dim E_{\text{residual}} = 15.$$

We call E_{residual} the *residual subspace*; it consists of those global $\lambda=12$ eigenvectors that are orthogonal to every zero-extension of a local $\lambda=12$ eigenfunction.

5.1 Descent of right multiplication to A_5 on $E_{V_{600}}(12)$

For every $h \in 2I$ let $P_h \in \text{Perm}(V_{600})$ be the 120-permutation matrix induced by right multiplication $v \mapsto vh$ on the vertex set. Right multiplication is an isometry of V_{600} , so P_h commutes with $L_{V_{600}}$ and restricts to a linear action on $E_{V_{600}}(12)$. Right multiplication on *vertices* has trivial kernel; descent to an A_5 action on a subspace $E \subset \mathbb{R}^{V_{600}}$ is the separate question of whether -1 acts trivially on E .

Lemma 4 ($\{\pm 1\}$ acts trivially on $E_{V_{600}}(12)$). P_{-1} restricted to $E_{V_{600}}(12)$ is the identity.

Proof by exact rational computation. Compute the $\mathbb{Q}$ -rational nullspace $B_{\text{global}} \in \mathbb{Q}^{120 \times 25}$ of $L_{V_{600}} - 12I$ (an integer matrix), giving a rational basis for $E_{V_{600}}(12)$. For each column v of B_{global} , apply the integer permutation matrix P_{-1} exactly and verify $P_{-1}v - v = 0 \in \mathbb{Q}^{120}$ componentwise (no floating point). Every one of the $25 \cdot 120 = 3000$ rational component checks returns 0. Hence $P_{-1} = \text{Id}$ on the basis, and therefore on $E_{V_{600}}(12)$. The computation is reproduced by `closure_transform_engine.keystone.exact_a5_characters` (field `minus_one_acts_trivially`). $\square$

By Lemma 4, the action of $2I$ on $E_{V_{600}}(12)$ factors through $A_5 = 2I/\{\pm 1\}$. To restrict this action to the summands E_{lifted} and E_{residual} we need $2I$ -invariance of each.

Lemma 5 (E_{lifted} and E_{residual} are $2I$ -invariant). For every $h \in 2I$, $P_h E_{\text{lifted}} = E_{\text{lifted}}$ and $P_h E_{\text{residual}} = E_{\text{residual}}$.

Proof. Right multiplication by $h \in 2I$ permutes the right cosets: P_h sends C_i bijectively to $C_{\sigma_h(i)}$, where $\sigma_h \in S_5$ is the induced coset permutation (Section 2.3). Restricted to a coset, $P_h|_{C_i}: C_i \rightarrow C_{\sigma_h(i)}$ is an isometry between two metric spaces that are each isomorphic to the 24-cell. It therefore intertwines the two local shell-1 Laplacians: $L_{C_{\sigma_h(i)}}(P_h v) = P_h(L_{C_i} v)$ for $v \in \mathbb{R}^{C_i}$. Consequently P_h maps $E_{C_i}(12)$ to $E_{C_{\sigma_h(i)}}(12)$, and by linearity of zero-extension, P_h maps $\text{lift}_i(E_{C_i}(12))$ to $\text{lift}_{\sigma_h(i)}(E_{C_{\sigma_h(i)}}(12))$. Summing over i gives $P_h E_{\text{lifted}} \subseteq E_{\text{lifted}}$; equality follows from invertibility of P_h . E_{residual} is the orthogonal complement of E_{lifted} in $E_{V_{600}}(12)$, and P_h is orthogonal (a permutation matrix), so P_h also preserves E_{residual} .

Independent of this prose proof, the repository carries an exact $\mathbb{Q}$ -rational certificate: for every $h \in 2I$ and for $B \in \{B_{E_{\text{lifted}}}, B_{E_{\text{residual}}}\}$, sympy computes $P_h B - B(B^\top B)^{-1} B^\top (P_h B)$ over $\mathbb{Q}$ and verifies that every component is exactly 0. This certifies $P_h B \subseteq \text{col}(B)$, hence P_h preserves E_{lifted} and E_{residual} , with no reference to the prose argument. Reproduced by the `lifted_2I_invariant` and `residual_2I_invariant` fields of `exact_a5_characters`. $\square$

By Lemmas 4–5, the A_5 action on $E_{V_{600}}(12)$ restricts to each of E_{lifted} and E_{residual} .

5.2 Exact A_5 class characters

Using rational bases of E_{lifted} , E_{residual} , and $E_{V_{600}}(12)$ constructed by sympy nullspace, we compute the trace of P_h on each subspace by exact rational matrix multiplication for every $h \in 2I$, and average by A_5 conjugacy class. The five A_5 conjugacy classes 1A, 2A, 3A, 5A, 5B are obtained by explicit orbit enumeration: we first enumerate the 60 distinct 5-letter permutations $\pi(h)$ for $h \in 2I$ (the image of the coset action), then partition them into orbits under conjugation $g \mapsto \pi g \pi^{-1}$ by all $\pi \in \pi(2I)$. The non-trivial cycle structures (1, 1, 1, 1, 1), (1, 2, 2) and (1, 1, 3) each give a single A_5 class (sizes 1, 15, 20); the cycle structure (5) splits into two A_5 orbits of size 12 each, which we label 5A and 5B. Class sizes (1, 15, 20, 12, 12) match the standard A_5 character table.

Theorem 6 (Exact A_5 class characters). The A_5 class characters of $E_{V_{600}}(12)$, E_{lifted} , and E_{residual} are constant on each conjugacy class and take exact integer values:

subspace	1A	2A	3A	5A	5B
$E_{V_{600}}(12)$	25	5	−5	0	0
E_{lifted}	10	2	−2	0	0
E_{residual}	15	3	−3	0	0

Each entry is computed in $\mathbb{Q}$ (no floating point).

Projecting the exact characters of Theorem 6 against the A_5 character table (irreps of dimensions 1, 3, 3, 4, 5; see [Conway et al. 1985]) gives:

irrep	dim	in E_{lifted}	in E_{residual}	in $E_{V_{600}}(12)$
Y_1	1	0	0	0
Y_3	3	0	0	0
Y'_3	3	0	0	0
Y_4	4	0	0	0
Y_5	5	2	3	5

Corollary 7 (A_5 irrep decomposition). *With Y_5 the 5-dimensional irreducible representation of A_5 ,*

$$E_{\text{lifted}} \cong 2 \cdot Y_5, \quad E_{\text{residual}} \cong 3 \cdot Y_5, \quad E_{V_{600}}(12) \cong 5 \cdot Y_5,$$

with every other complex A_5 irrep having multiplicity exactly 0 (the character inner products that compute these multiplicities are themselves computed in $\mathbb{Q}(\sqrt{5})$ using the exact character-table entries).

The “15 = 3 × 5” is representation-theoretic — three copies of the 5-dim A_5 irrep — *not* five cosets each contributing three local modes.

6 Interpretation and limits

This is a reproducible spectral-graph result in the 24-cell/600-cell coset geometry. It does not by itself derive particle physics, quantum mechanics, or cosmology.

Within the broader closure-geometry programme that motivated this work, the result may be interpreted as one example of a “closure-projection channel”: a local shell structure lifting exactly into a global invariant sector. That interpretation is provided here only for context and is not load-bearing for any claim in this note.

7 Reproducibility

One-command reproduction. From the repository root:

```
python closure_transform_engine/examples/run_wo008_keystone.py
pytest closure_transform_engine/tests/test_wo008_keystone_artifact.py
```

The first command (a few minutes on commodity hardware) builds the 600-cell from exact $\mathbb{Q}(\sqrt{5})$ icosian quaternions, runs all certificates, and writes the frozen output artifacts listed below; the second re-asserts the certificate-level claims listed below as a test.

Environment. The repository pins the toolchain in `requirements.txt` and `pyproject.toml`: Python ≥ 3.8 (developed on 3.8.10), NumPy ≥ 1.20 (1.24.4), SymPy ≥ 1.12 (1.13.3), SciPy ≥ 1.7 (1.10.1). The formal certificates use only exact $\mathbb{Q}$ arithmetic (SymPy Rational/Matrix); NumPy enters only the floating-point cross-check of Remark 2. The `wo008_reproducibility_log.txt` output records the run UTC timestamp, elapsed time, Python version, repository path, and the repository’s commit hash (with an “uncommitted changes present” marker if the working tree was dirty at run time), so the artifact set is anchored to a specific source revision.

Repository contents (proof package). The repository is part of the proof package, not an afterthought. It contains: the exact construction of V_{600} , $2T$, the five cosets, and the $d=1$ graph Laplacians; the exact nullspace and certificate routines for Theorem 1, Lemma 4, Theorem 6, and the integer spectrum; the A_5 conjugacy-class enumeration and character projection; the test suite; and the frozen output artifacts (`wo008_summary.json`, `wo008_verdict_table.csv`, `wo008_reproducibility_log.txt`). The companion scripts cited from [Paper V] and [Paper XXXV] live in the main VFD research repository (paths given in the bibliography); the self-contained reproduction bundle for this note vendors only the `vfd_v600` icosian package it depends on. *The claims of this note do not require rerunning Paper V or Paper XXXV.* Those upstream references justify the broader programme context and provide an independent treatment of the coset construction; the standalone bundle reconstructs from scratch the objects needed for the certificates used here, so a reader can verify every theorem of this note without access to the main research repository.

What the runner verifies:

- Theorem 1 via the $\mathbb{Q}$ -rational certificate (integer matrices, $\mathbb{Q}$ -rational nullspace bases, exact zero off-coset components after lift in every one of $5 \cdot 2 = 10$ local basis vectors).
- The floating-point cross-check of Remark 2 (max residual $\leq 10^{-10}$; observed $\approx 6.2 \times 10^{-15}$).
- Every integer-eigenvalue multiplicity of L_C and $L_{V_{600}}$ listed in the spectra (exact $\mathbb{Q}$ -rational sympy nullspace at each integer λ).
- The intersection-dimension table of Section 4.
- Lemma 4 ($-1 \in 2I$ acts trivially on $E_{V_{600}}(12)$) by exact $\mathbb{Q}$ -rational basis check.
- Lemma 5 ($2I$ -invariance of E_{lifted} and E_{residual}): exact $\mathbb{Q}$ -rational check of $P_h B = B(B^\top B)^{-1} B^\top (P_h B)$ for every $h \in 2I$ and both $B = B_{E_{\text{lifted}}}, B_{E_{\text{residual}}}$.
- Theorem 6 (exact integer A_5 class characters) and Corollary 7 (A_5 irrep multiplicities) by exact rational trace and character projection.

Outputs:

- `wo008_summary.json` — full structured result including the certificate’s per-coset status.
- `wo008_verdict_table.csv`
- `wo008_reproducibility_log.txt`

References

- [Coxeter 1973] H. S. M. Coxeter, *Regular Polytopes*, 3rd ed., Dover, 1973.
- [Conway–Sloane 1999] J. H. Conway and N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, 3rd ed., Springer-Verlag, 1999.
- [Conway et al. 1985] J. H. Conway, R. T. Curtis, S. P. Norton, R. A. Parker and R. A. Wilson, *Atlas of Finite Groups*, Oxford University Press, 1985.
- [Paper V] *Polytope-uniqueness theorem and spectra.* In the main VFD research repository: `papers/paper-v/paper-v.tex`; supporting scripts `papers/paper-v/scripts/run_polytope_uniqueness_narrow.py` and `papers/paper-v/scripts/run_rigorous_verification.py`.

- [Paper XXXV] *Schläfli decomposition of the 600-cell and the A_5 origin of the five inscribed 24-cells.* In this repository: `papers/paper-xxxv/paper-xxxv.tex`; supporting scripts at `papers/cascade-derivation/scripts/verify_schlafl_i_via_2T.py` and `papers/cascade-derivation/scripts/coset_action_2I_on_5_skeletons.py`. Paper XXXV proves the left-coset version explicitly; the right-coset version used here follows from the involution $g \mapsto g^{-1}$ of $2I$, which exchanges the left- and right-coset families and preserves their common identity coset.
- [V₆₀₀-programme] *Exact $\mathbb{Q}(\sqrt{5})$ icosian arithmetic and $2I/2T/\sigma$ -Galois machinery.* In this repository: `papers/v600-programme/lib/vfd_v600/`.